Quick Start Guide
Picosecond Pulsed Fiber-Coupled Laser
PLD-XXX-SM-FC-30-U



1 About This Guide
This document provides a concise overview for installation and basic operation of the device. It does **not** replace the full product manual.

The complete manual is available at: https://www.flimlabs.com/products/pulsed-laser-diode/

Scan the QR code to access the full manual.

Warning Read the full manual before operating the device. Improper use may result in hazardous laser exposure.

2 Safety Information
This product is classified as a Class 3R laser according to IEC EN CEI 60825-1 (ed. 3, 2014) and may emit radiation exceeding the maximum permissible eye exposure (MPE), posing a risk of serious eye injury when the beam is viewed directly.

Warning Visible and invisible laser radiation. Avoid direct eye exposure.

- Never look into the fiber output.
- Do not operate the device without a connected optical path.
- Do not point the laser at people or reflective surfaces.
- Operate only within the specified electrical and environmental limits.
- Do not open the enclosure.

3 Intended Use
The Picosecond Pulsed Fiber-Coupled Laser is intended for laboratory and research applications requiring pulsed or continuous laser excitation, such as fluorescence spectroscopy and FLIM measurements.

The device must be operated only by qualified personnel.

4 Product Overview
This laser is a compact fiber-coupled diode laser capable of continuous (CW) and pulsed operation. Different models are available each emitting at a different wavelength as listed in the table below.

Key Features

- Multiple wavelengths (405–850 nm)
- Fiber-coupled single-mode output
- CW and pulsed operation (5–80 MHz)
- External synchronization (SYNC IN)
- TTL synchronization outputs

Model	Wavelength
Visible radiation:	
PLD-405-SM-FC-30-U	405 nm
PLD-445-SM-FC-30-U	445 nm
PLD-488-SM-FC-30-U	488 nm
PLD-520-SM-FC-30-U	520 nm
PLD-530-SM-FC-30-U	530 nm
PLD-635-SM-FC-30-U	635 nm
Invisible radiation:	
PLD-808-SM-FC-30-U	808 nm
PLD-850-SM-FC-30-U	850 nm

5 Components Overview
Front panel



1. FC/PC output fiber connector
2. ON/OFF LED
3. Power selector (%): 20; 30; 40; 50; 55; 60; 65; 70; 75; 80; 90; 100
4. Operational mode selector: Off; External Synchronization; Repetition Frequency; CW

Rear panel



1. Remote interlock
2. Power input (USB Type-C)
3. Status LED (POWER GOOD)
4. SYNC OUT 1 – synchronization output port 1
5. SYNC OUT 2 – synchronization output port 2
6. SYNC IN – synchronization input port

6 Scope of Delivery
- PLD laser module
- Optical fiber (single-mode)
- SMA cable (SYNC)
- USB Type-C power cable
- Documentation

7 Setup
Place the device on a stable, vibration-free laboratory surface with sufficient space for cable routing.

Before connecting the device, verify the following:

- The device and all cables are free of visible damage
- The optical fiber is clean and undamaged
- A suitable power source is available (5 V DC via USB Type-C, ≥ 3 A)

8 Connections
All connections must be completed before enabling the laser emission.

Power Connection

- Set the operating mode selector to **OFF**
- Connect the USB Type-C cable to the device
- Connect the power supply and verify that the power LED is illuminated

Optical Connection

- Connect the optical fiber to the FC/PC output
- Ensure that the fiber is properly seated and not under mechanical stress
- Always ensure a closed optical path before enabling the laser

Synchronization Interfaces

- SYNC IN: accepts an external TTL signal (2.5 V, 50 Ω) to trigger the laser
- SYNC OUT: provides a TTL signal synchronized with the laser emission

9 Operation
After completing all connections, the device is ready for operation. The laser can be operated in continuous or pulsed modes using the front panel controls.

Continuous Mode (CW)

- Set the operating mode selector to **CW**
- Adjust the output power using the power selector

Pulsed Mode (Internal Trigger)

- Select the desired repetition rate (5–80 MHz)
- Adjust the output power as required

External Trigger Mode

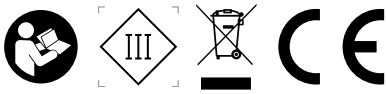
- Set the operating mode selector to **EXT SYNC IN**
- Apply an external TTL signal to the SYNC IN port
- The laser emission follows the external trigger signal

Warning Enable the laser only after all connections are correctly established and verified.

10 Further Information
Detailed instructions, safety information, and troubleshooting are available in the full manual: https://www.flimlabs.com/products/pulsed-laser-diode/

11 Product Label

Picosecond Pulsed Fiber-Coupled Laser
P/N: PLD-405-SM-FC-30-U
S/N: PLD-XXXX-XXX
5 V ⎓ 3.0 A | IP40



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12 Technical Specifications

Parameter	Value
Electrical	
Electrical safety class	Class III
Power supply	5 V DC (USB Type-C)
Maximum current consumption	3 A
Maximum power consumption	15 W
Optical and Laser	
Laser type	Solid-state diode laser
Available wavelengths	405, 445, 488, 520, 530, 635, 808, 850 nm
Wavelength bandwidth	±10 nm
Operating modes	Continuous (CW) or pulsed
Maximum output power	1.4 mW
Power stability	< 1%
Pulse repetition frequency	5, 10, 20, 40, 80 MHz
Pulse duration	50, 55, 80, 80, 90, 80, 90, 90 ps
Output fiber core diameter	≥ 4 µm
M^2 value	1.06 (with 4 µm core fiber)
Polarization	Random
Optical output	FC/PC fiber connector
Signal	
Manual command mode	Selectable repetition frequency
External synchronization input	TTL, 2.5 V, 50 Ω
Synchronization outputs	TTL, 2.5 V, 50 Ω
Physical	
Dimensions (L x W x H)	135 mm x 110 mm x 50 mm
Weight	550 g
Environmental	
Operating temperature	10°C to +40°C
Storage temperature	-20°C to +70°C
Relative humidity	Up to 80% non-condensing

Table 1: Key specifications of the Picosecond Pulsed Fiber-Coupled Laser.

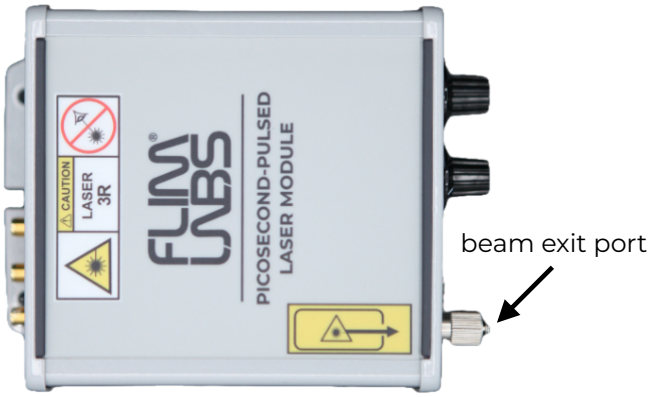


Figure 1: Top view of the laser with indicated output.